

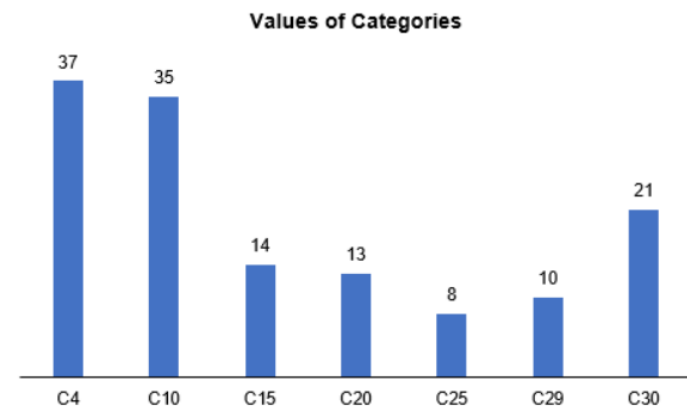
DIRECTIONS for questions 1 to 5: Answer these questions on the basis of the information given below.

During the month of April, Harish, the headmaster of a residential school, tracked the attendance of the students in his school. The students are expected to attend school on all the 30 days of the month. Each student attended the school on at least one day during April.

Harish created different categories, C1, C2, ... C30, representing the number of consecutive days (from 1, 2, ... 30, respectively) for which a student attended the school during the month. For each student, Harish counted all the numbers of consecutive days that the student attended school during the month. He then increased the value of all the applicable categories for the applicable number of times. Further, the student is said to have *contributed* to a category if his attendance has contributed to an increase in the value of that category.

For example, if a student attended the school from April 3rd to April 6th (i.e., for four days), then from April 7th to April 10th (again for four days), and then from April 20th to April 30th (for eleven days) and not on any other days, Harish would increase the value of C4 by two and C11 by one. Further, the student is said to have *contributed* twice to C4 and once to C11.

The graph below provides the values of all the categories for which the value is greater than 0, considering the attendance of all the students for April.



Q1. DIRECTIONS *for question 1:* Select the correct alternative from the given choices.

If no student *contributed* to more than one category, what is the minimum number of students in the school?

- ☐ a) 108
- ☒ b) 91 ✓ **Your answer is correct**
- ☐ c) 98
- ☐ d) 90

Q2. DIRECTIONS *for question 2: Type in your answer in the input box provided below the question.*

What is the minimum possible number of students in the school?

Q3. DIRECTIONS *for questions 3 and 4:* Select the correct alternative from the given choices.

If the number of students in the school is the minimum possible, what is the minimum number of days that any student could have attended the school during the given month?

☒ a) 4 ✖ **Your answer is incorrect**

☐ b) 8

☐ c) 10

☐ d) 20

Q4. DIRECTIONS *for questions 3 and 4:* Select the correct alternative from the given choices.

What is the maximum number of students who could have attended the school for exactly 24 days during the month?

- ☐ a) 30
- ☒ b) 13 **✗ Your answer is incorrect**
- ☐ c) 37
- ☐ d) 31

Q5. DIRECTIONS *for question 5:* Type in your answer in the input box provided below the question.

If the number of students in the school is the minimum possible, what is the maximum number of students who could have attended the school for exactly 24 days during the month?

DIRECTIONS for questions 6 to 10: Answer these questions on the basis of the information given below.

Meridian Business Academy, an MBA college, conducted an exam in which there were exactly 20 questions of multiple-choice type, such that each question had four options labelled A, B, C and D and the correct answer to each question was exactly one of these four options. Each question answered correctly was awarded one mark and there were no negative marks for any questions answered incorrectly or left unattempted.

After the exam, the college decided to calculate the scores of all the students who wrote the exam using a software, which compares the correct answer to each question with the response of each student for that question, and calculates the total marks of each student. However, due to some error, for all the questions for which the correct answer was option A, the software evaluated the responses considering option B as correct; for all the questions for which the correct answer was option B, the software evaluated the responses considering option C as correct; for all the questions for which the correct answer was option C, the software evaluated the responses considering option D as correct; and for all the questions for which the correct answer was option D, the software evaluated the responses considering option A as correct.

The college initially published the scores of all the students as calculated by the software, without realizing this error. Upon receiving numerous complaints, the college reprocessed the results (correctly) and published a revised score for each student.

Q6. DIRECTIONS for questions 6 and 7: Select the correct alternative from the given choices.

If a student received a score of 8 initially, what is the maximum revised score that he could have received in the exam?

- ☐ a) 8
- ☐ b) 10
- ☐ c) 12
- ☐ d) 20

Q7. DIRECTIONS *for questions 6 and 7:* Select the correct alternative from the given choices.

P, Q, R and S marked the answer to all the questions as option A, B, C and D respectively. If P, Q and R initially received a score of 5, 3 and 4 respectively, what is the difference between the initial score and revised score of S?

☐ a) 5

☐ b) 3

☐ c) 4

☐ d) 2

Q8. DIRECTIONS *for question 8:* Type in your answer in the input box provided below the question.

P, Q, R and S marked the answer to all the questions as A, B, C and D respectively. The initial scores of P, Q and R were more than their revised scores by 3, 3 and 1 respectively.

What is the revised score of S?

Q9. DIRECTIONS *for question 9:* Select the correct alternative from the given choices.

P, Q, R and S marked the answer to all the questions as A, B, C and D respectively. The initial scores of P, Q and R were more than their revised scores by 3, 3 and 1 respectively.

For how many questions is the correct answer to the question option B?

☐ a) 1

☐ b) 2

☐ c) 3

☐ d) 4

Q10. DIRECTIONS *for question 10:* Type in your answer in the input box provided below the question.

P, Q, R and S marked the answer to all the questions as A, B, C and D respectively. The initial scores of P, Q and R were more than their revised scores by 3, 3 and 1 respectively.

If a student marked the answer to the first ten questions as A and the next ten questions as D, what is the maximum possible difference between his initial score and revised score?

DIRECTIONS *for questions 11 to 15: Answer these questions on the basis of the information given below.*

Each of five seminars – *Motorcycle Maintenance (MM)*, *Sedan Service (SS)*, *Truck Tune-up (TT)*, *Rickshaw Repair (RR)* and *Oxcart Overhaul (OO)* – was conducted on each of on five days, from Day 1 to Day 5. The five seminars were conducted by five persons among Ravi, Satish, Tarun, Umar and Varun, not necessarily in that order, such that each seminar was conducted by the same person on any day. On any day, all the five seminars started at the same time and ended at the same time.

Five persons – Amar, Binoy, Charan, Dev and Emran – attended the five seminars on the five days such that, on any day, each seminar was attended by exactly one of the five persons. Further, each seminar was attended exactly once by each of the five persons.

Further, it is also known that

- i. Binoy attended the seminar conducted by Tarun on Day 4.
- ii. Charan attended SS before Amar did.
- iii. the seminar that Dev attended on Day 3 was the same as that attended by Emran on Day 5 and this seminar was conducted by Satish.
- iv. Amar attended OO before either Binoy or Charan attended OO.
- v. Binoy attended TT before either Amar or Charan attended TT.

vi. Charan attended RR before either Amar or Binoy attended RR.

vii. Emran attended TT, SS and MM on three consecutive days, in that order, and he did not attend TT on Day 1.

viii. Charan attended TT and OO on consecutive days in that order.

ix. Dev attended MM on Day 1, which was conducted by Umar.

x. Binoy attended RR after Dev did.

Q11. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.

Which of the following BEST represents the seminar that Charan attended on Day 5?

- ☐ a) **TT**
- ☐ b) **MM**
- ☐ c) **OO**
- ☐ d) Either TT or MM

Q12. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.

Which of the following BEST represents the day(s) on which Dev could have possibly attended the seminar conducted by Ravi?

☐ a) **Day 2 or Day 4 or Day 5**

☐ b) **Day 4**

☐ c) **Day 2 or Day 5**

☐ d) **Day 4 or Day 5**

Q13. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.

Who among the following attended the seminar conducted by Tarun on Day 2?

- ☐ a) **Amar**
- ☐ b) **Emran**
- ☐ c) **Dev**
- ☐ d) Charan

Q14. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.

How many of the five persons attended SS before Binoy did?

- ☐ a) 3
- ☐ b) 2
- ☐ c) 4
- ☐ d) Either 3 or 4

Q15. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.

How many of the five seminars did Amar and Dev attend on consecutive days, in any order?

☐ a) 0

☐ b) 2

☐ c) 3

☐ d) 4

DIRECTIONS *for questions 16 to 20:* Answer these questions on the basis of the information given below.

In a college, each course is worth a certain number of credits. Tarun, a student in this college, has to select courses from CM, IDF, ME, OB, OD, OM and SM. He has to select the courses such that the total credits that he earns from the courses that he selects is at least 12. It is known that SM, IDF and ME are worth three credits each, CM and OB are worth four credits each and OM and OD are worth five credits each. Further, it is also known that

- i. if he selects CM, he must select IDF.
- ii. if he selects IDF, he must select other courses such that the number of credits that he earns is not more than 16.
- iii. between OM, CM and SM, he must select exactly two courses.
- iv. he must either select both OD and ME or select none of the two courses.
- v. he has to select at least one course worth 3 credits, at least one course worth 4 credits and at least one course worth 5 credits.

Q16. DIRECTIONS *for questions 16 and 17:* Type in your answer in the input box provided below the question.

In how many ways can Tarun select his courses?

Q17. DIRECTIONS *for questions 16 and 17:* Type in your answer in the input box provided below the question.

What is the highest possible number of credits that Tarun can earn?

Q18. DIRECTIONS *for questions 18 to 20:* Select the correct alternative from the given choices.

If Tarun wants to do the minimum possible number of courses, which of the following courses must Tarun select?

☐ a) **OM**

☐ b) **CM**

☐ c) **OB**

☐ d) ME

Q19. DIRECTIONS *for questions 18 to 20:* Select the correct alternative from the given choices.

If Tarun wants to select ME, what is the minimum number of courses that he must select, not counting ME?

☐ a) 2

☐ b) 3

☐ c) 4

☐ d) 5

Q20. DIRECTIONS *for questions 18 to 20:* Select the correct alternative from the given choices.

If Tarun can select exactly one course among OM and OB, what is the maximum number of courses that he can select?

☐ a) 3

☐ b) 4

☐ c) 5

☐ d) 6